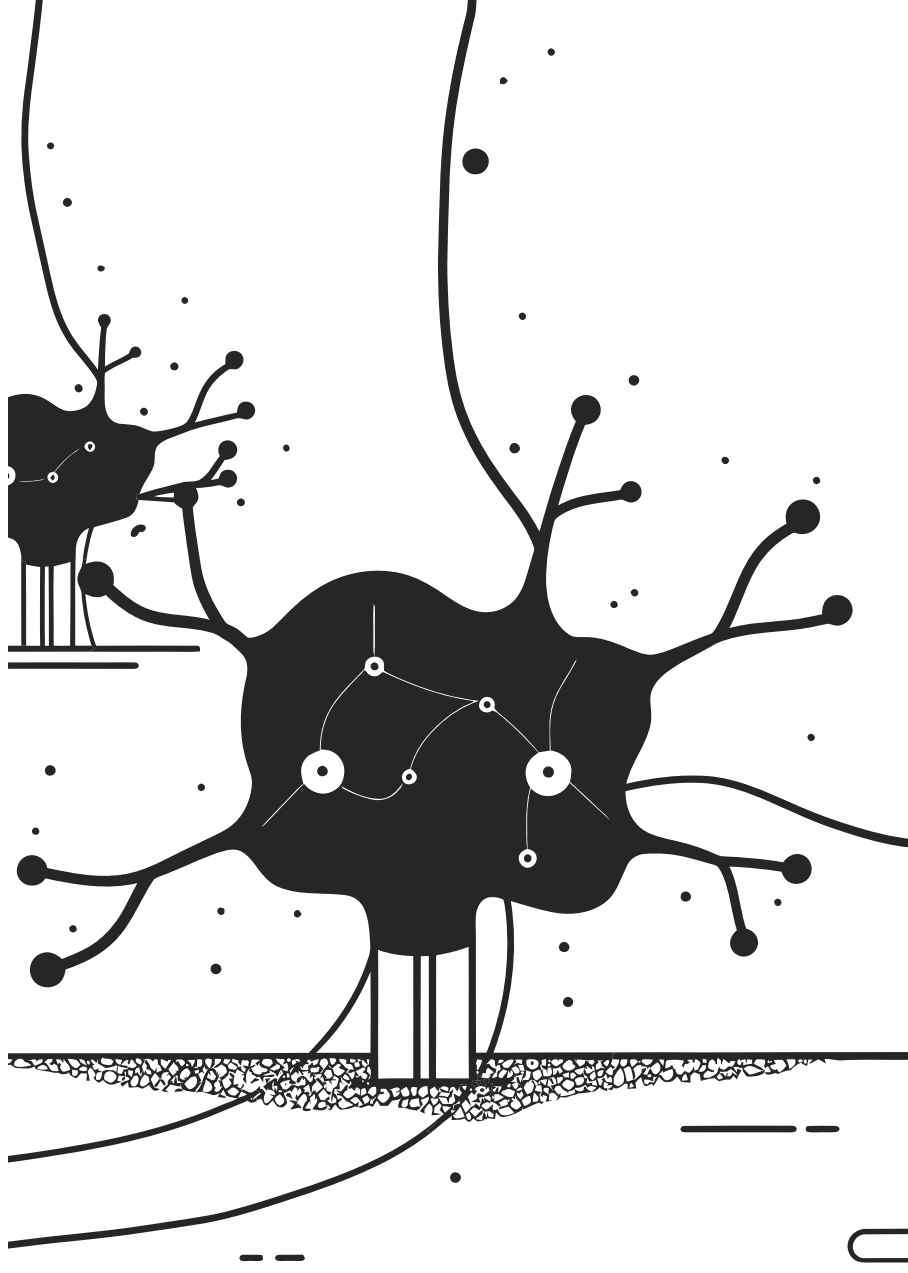
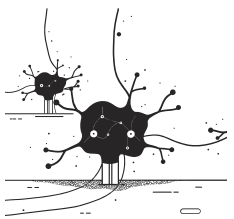
How Blockchain Produces Integrity

How a record becomes tamper-evident without anyone in charge.

SOURCE: DRESCHER (2026), BLOCKCHAIN BASICS (2ND ED.), STEPS 4-13

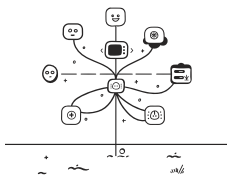


The Core Ideas



Hash

A fixed-size fingerprint of data. The same input always gives the same output - any change, however small, changes the output completely.



Merkle Tree

A hash summary of many items, allowing efficient and verifiable proof that any single item belongs to a set.



Digital Signature

Cryptographic proof of authorisation - proves a change was made by the holder of a private key.



Double-Spend

Using the same coin twice. Hashing alone does not prevent this - consensus does.

What Hashing Does - and Does Not Do

What it does

Tamper-evident links

Each block holds the previous block's hash. Altering any old record breaks every later link - the change is immediately visible.

Proof of authorisation

Combined with digital signatures, hashing gives tamper-evident integrity with no trusted third party required.

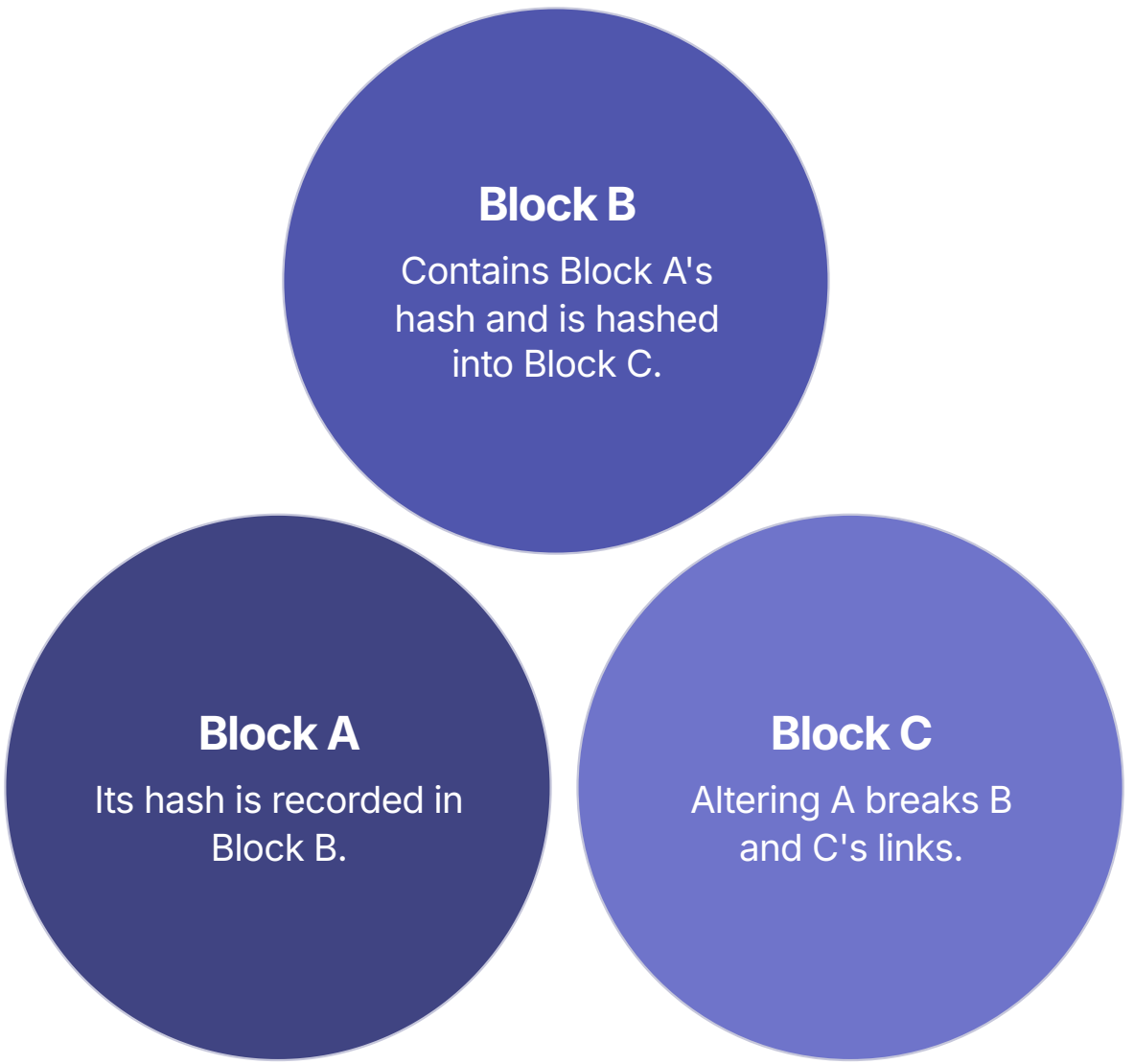
What it does not do

Does not keep data private

Hashing makes records verifiable, not hidden. Anyone with access can read the data.

Does not stop double-spending

Preventing duplicate use of a coin requires consensus across the network - hashing alone is not enough.



Altering one block cascades - every subsequent link breaks, making tampering detectable across the entire chain.

Worked Example: The Bank Auditor

> "Can we prove these records were not altered?"

Hashing answers **yes** - any change to a record breaks the hash chain and becomes immediately visible.

What hashing proves

The records have not changed since they were written.

What hashing does not prove

That the records are correct, or that their contents are private.

